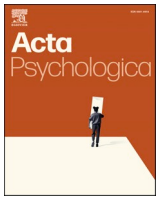
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Digital literacy and technostress as dual pathways linking work-from-home practices to employee performance: Evidence from the post-digital workplace

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ABSTRACT

The primary objective of this study was to assess employee performance among individuals working remotely, considering the mediating role of digital literacy and the moderating influence of technostress within public universities in Malaysia. A structured questionnaire was used to gather 320 responses from university lecturers. The data were analyzed using the partial least squares structural equation modelling (PLS-SEM) technique through smartPLS-4. The findings indicate that work-from-home (WFH) does not exhibit a direct significant relationship with employee performance, except for an indirect relationship mediated by digital literacy. Conversely, technostress significantly and negatively moderated the relationship between digital literacy and employee performance. Moreover, technostress moderated the indirect relationship between WFH and employee performance through digital literacy. Consequently, digital literacy is essential for the successful implementation of effective work-from-home strategies that enhance employee performance and well-being at work. University management should prioritize enhancing digital literacy before implementing remote work strategies. Additionally, management should address employee technostress levels, as it can influence both digital literacy and employee performance. Future research should explore these findings in other service sectors, such as information technology and customer services.

1. Introduction

In March 2020, most countries, including Malaysia, implemented lockdowns to control the spread of Covid-19. These lockdowns enforced restrictions on movement, international travel, and the closure of industries, businesses, government offices, and educational institutions (Shah et al., 2020). In response to the lockdown, most companies and educational institutions forced their employees to work from home (WFH) to accomplish their daily tasks (Almeida et al., 2024). Almost two years later, most governments opened their markets and resumed work as usual. However, the pandemic has left a permanent mark on the principles of the workplace, as many companies' management and their employees prefer to continue working from home, especially in the education sector (Anik & Habib, 2025; Yeravdekar et al., 2024). Therefore, it is reasonable to assume that work from home (WFH) is a permanent practice (Smite et al., 2023).

In addition, WFH benefits disabled employees and those caring for dependents. Furthermore, from an employer's perspective, it can reduce operational costs, such as workspace, parking, and facilities (Shafizadeh et al., 2007). The pandemic has transformed remote working from a rare occurrence for many firms and their employees into a new norm (Yao et al., 2023). This study shows how the WFH strategy changes employees' performance. The WFH strategy changes the working environment, which is a challenge for the management of employee performance (Saleem et al., 2021).

Thus, this study aims to explore a better understanding of WFH and its impact on employee performance in public universities in a developing country from a post-pandemic perspective. Additionally, this study seeks to clarify how digital literacy and technostress influence the connection between WFH and employee performance, filling a gap in the existing research.

The next section covers the literature and proposes hypotheses. The

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third and fourth sections cover the methodology and data analysis of the study, respectively. The final section discusses the conclusion, implications, limitations, and future recommendations.

2. Employee performance

It is very difficult to quantify employee performance because of the multiple tasks and their daily role on the job. From some perspectives, it is described as the completion of tasks under given job descriptions (Pawar, 2013). In other words, employee performance can also be explained as the difference between excellent and poor-performing employees and how actively they perform or complete their tasks on time or within the deadline (Lee, 2010). Consequently, employee performance can significantly influence an organization's overall performance (Okolie & Kawedo, 2018).

Moreover, past studies have operationalized employee performance differently. For instance, the first group of researchers defined employee performance as an individual's total output (Sonnentag et al., 2008). The second group stated that they completed tasks under the given job description (Darsana, 2014). The third and last group identified positive behavior towards extra role performance, which can have significant implications for overall firm performance (Motowidlo, 2013). In addition, one study explained employee performance as the total number of quantitative and qualitative contributions of an individual or group to the overall performance of the firm (Sen & Dulara, 2018).

Aside from the operational definition of employee performance during or after a pandemic, more organizations are giving their employees the freedom to work in the office or from home as they choose. According to Patanjali and Bhatta (2025), WFH not only increases employee productivity but also saves companies' expenses by reducing the use of offices and utilities. Apart from WFH, technostress and lack of digital literacy are also considered crucial factors for employee performance in a telecommuting environment, which can negatively affect the employee performance (Khorakian et al., 2024). Thus, this study investigates the influence of WFH, digital literacy, and technostress on employee performance to draw general conclusions.

2.1. Work from home (WFH) and employee performance

WFH refers to a work arrangement in which employees perform their job duties remotely, typically from their homes or any location outside the traditional office environment (Pullokaran & Joseph, 2023). Instead of commuting to a physical workplace, employees use digital technologies and communication tools to connect with colleagues, access work-related resources, and fulfill their responsibilities (Lal et al., 2021). Generally, employees will be operated remotely through telecommuting and virtual work environments where a physical body is not required, which is the opposite of working in a traditional way. The same idea behind remote work is applied to working from home. Many employees work from home but still come to the office. The term “remote work” refers to a broader set with four components: work location, which can be anywhere; diversity of employment relationships; time distribution; and the use of ICT.

The focus of WFH research is on how the idea affects employee performance (Jalagat & Jalagat, 2019). WFH is also known as the intra-firm decentralization of power, which can improve employees' freedom and influence over the tasks, pace, and location of their jobs (McAlpine et al., 2024). Angelici and Profeta (2020) conducted their research within the context of a large company in Italy and found that working with flexibility (both in terms of location and time) boosts employee performance and wellness. This result is in marked contrast to employers' concerns that workers would perform poorly because of a lack of accountability while WFH.

Furthermore, WFH provides employees with greater flexibility and autonomy in managing their work schedules than work from office. This flexibility enables individuals to work during their most productive hours, balance personal and professional responsibilities, and create a conducive work environment tailored to their preferences (Yu & Wu, 2021). When employees have control over their work, it enhances their motivation, job satisfaction, and overall performance (van der Kolk et al., 2019). However, physical distance may pose challenges for collaboration, and advancements in digital communication tools and technologies have made remote collaboration more feasible (Pallot, 2011). Video conferencing, instant messaging, and project management tools enable employees to connect and collaborate with colleagues effectively in the WFH mode (Rysavy & Michalak, 2020).

Moreover, WFH can contribute to an improved work-life balance, as employees have the flexibility to integrate personal and professional commitments more seamlessly. Achieving a better balance between work and personal life can enhance well-being, reduce stress, and ultimately have a positive impact on performance. According to data from the British, employees working from home during the pandemic feel more motivated and independent (Pelly et al., 2021). One more study stated that employees who WFH can improve communication with their organization by using digital tools extensively (Hauret et al., 2020). A study conducted in Germany during the pandemic found that WFH could increase employees' motivation and performance (Kifor et al., 2022). Based on the above discussion, the following hypothesis is proposed:

H1. WFH positively affects employee performance in public universities in Malaysia.

2.2. Mediating role of digital literacy

Digital literacy refers to the ability to use and navigate digital technologies effectively and responsibly. It encompasses a range of skills and competencies that enable individuals to find, evaluate, create and communicate information using digital devices and online platforms (Fraillon et al., 2014). Digital literacy can be referred to as the fundamental organizational and physical structures required for the functioning of a society or business, as well as the services and amenities required for an economy to run smoothly (Hanseth & Lyytinen, 2016). However, some key components of digital literacy, which are important for achieving affective digital literacy, include basic computer skills, Internet skills, information literacy, communication and collaboration, digital security and privacy, and media literacy (Niu et al., 2024). Digital literacy is important in today's digital world because it empowers individuals to fully participate in the digital society, access information, communicate effectively, and make informed decisions (Sharma et al., 2016). Thus, these skills are essential for academic staff to acquire new knowledge, secure employment, engage in civic activities and pursue personal development.

However, digital skills are crucial for employees to WFH (Sadik Tatli et al., 2023). In other words, lecturers with strong digital literacy skills can quickly adapt to remote teaching platforms, online collaboration tools, and virtual communication channels. They can efficiently use digital resources, manage online classes, and engage students effectively. This adaptability can positively impact their overall performance as lecturers (Moorhouse & Wong, 2022). Moreover, WFH heavily relies on digital tools and technologies; lecturers with high digital literacy can utilize these tools efficiently, such as learning management systems, video conferencing platforms and online assessment tools. They can effectively deliver lectures, share resources, interact with students, and provide timely feedback (Hafiz & Fitria, 2022). Proficient use of digital tools enhances productivity and can positively influence lecturer performance.

Thus, digital literacy skills aid in efficient time and task management. Lecturers can organize their work schedules, create digital calendars, set reminders, and prioritize tasks. They can also leverage productivity tools and project management platforms to streamline workflows. Effective time and task management contribute to increased productivity and improved performance. Based on this, the current

study proposes the following hypothesis:

H2. Digital literacy positively mediates the relationship between WFH and employee performance in public universities in Malaysia.

2.3. Moderating role of technostress

Technostress refers to the psychological and physiological strain or discomfort experienced by individuals due to their interaction with technology (Salanova et al., 2013). It arises when individuals perceive that the demands and pressures associated with technology use exceed their ability to cope (Finstad et al., 2024). Technostress can manifest in various ways and impact different aspects of well-being and performance. Another study stated that technostress was expanded to include "any adverse effect induced directly or indirectly by technology on attitudes, thoughts, behaviors, or psychology" (Shu et al., 2011).

Furthermore, technology users experienced psychological pressure, which is distinguished by displeasure and dissatisfaction because of the times brought on by technology, which are changing too quickly to accommodate individual locations (Taylor, 2022). Anxiety, mental weariness, sadness, and nightmares are among the symptoms of technostress; nevertheless, many people also have frequent rage attacks driven by the challenges of using computer software and managing errors or roadblocks that prevent them from finishing their tasks (Weinert et al., 2020). Moreover, information overload, constant connectivity, technology complexity, lack of control, digital distractions, and fear of technology obsolescence are the causes of technostress (Haque et al., 2019). Therefore, technostress negatively affects individual mental health and overall employee performance.

Regarding the moderating role of technostress, high levels of digital literacy can help lecturers cope with technostress (Haque et al., 2021). With strong digital literacy skills, lecturers are better equipped to handle technological challenges, troubleshoot issues and find alternative solutions. This can reduce the negative impact of technostress on employee performance (Çınar & Kenek, 2025). Conversely, low digital literacy can exacerbate technostress. Lecturers who lack digital literacy skills may struggle to navigate digital tools, feel overwhelmed by technological demands, and experience increased technostress (Kurniawan et al., 2024). This can further hinder their performances.

However, the degree of fit between lecturers' digital literacy skills and the technological demands of their work can influence the impact of technostress on their performance. Lecturers with a high level of digital literacy that aligns well with the technology they use may experience less technostress and achieve better performance (Bartra-Rivero et al., 2024). Therefore, this study proposes the following hypothesis:

H3. The moderating role of technostress negatively affects the relationship between digital literacy and employee performance in Malaysian public universities.

In addition, this study conceptualized the mediating role of digital literacy between WFH and employee performance; this view motivated the present study to investigate the effect of technostress on the indirect relationship between WFH and employee performance through digital literacy. Based on this, the following hypothesis is proposed:

H4. Technostress moderates the indirect relationship between WFH and employee performance through digital literacy, whereby this indirect relationship is stronger for employees experiencing lower technostress and weaker for those experiencing higher technostress.

2.4. Theoretical grounds for research work

This study covers the transaction model of stress and coping (TMSC). This model has been used in many studies in the fields of human resources, education, health, and sports (Lim et al., 2023). Lazarus and Folkman (1984) explained the TMSC model that "an individual's appraisal of environmental stimuli as being stressful will lead to coping strategies, which consequently will compel psychological changes within the individual as an outcome of coping with stress." In other words, it refers to how individuals perceive and respond to stressful transitions. Further, TMSC conceptualizes "stress as a product of dynamic interaction between an individual and their environment (Lazarus & Folkman, 1984)." In short, an individual can manage stressful situations through effective appraisal and coping strategies. Furthermore, appraisal is divided into primary appraisal, which evaluates stress, and secondary appraisal, which assesses the controllability of perceived stressors (Kathukumi, 2023).

TMSC links with this study work on how university lecturers evaluate the WFH and related technological stressors, how they appraise these stresses, and how they can use their available resources to cope with these stresses effectively to maintain their outcome, such as higher performance (Hamza Shuja et al., 2020; Lazarus & Folkman, 1984; Miyata et al., 2015). The implementation of the WFH policy in universities could be a potential stressor for lecturers; hence, this is covered under the TMSC model to evaluate the stresses due to WFH and technostress, implement coping strategies to handle the stresses such as digital literacy, and control the overall outcomes such as diffusion of stress, which leads to higher employee performance (Zaza & Erskine, 2024).

Thus, WFH and technostress are considered external environments that can be stressful events. On the other hand, individual evaluation to cope with the stressor event, such as in our study case, the secondary appraisal component, can be the digital literacy skill, which further can control the stresses and improve the employee performance in the framework. The proposed connection between the variables under the TMSC theory is shown in Fig. 1.

3. Research methods

This study focuses on academic staff (lecturers) employed at Malaysian public universities. According to Sekaran and Bougie (2016), a sample size of 30–500 is appropriate for social science studies. However, the present study collected 320 responses from academic staff (lecturers) at public universities in Malaysia. In this study, the unit of analysis consisted of individual respondents. Moreover, the convenience sampling technique was used to approach the lecturers engaged in online classes through their official e-mails, contact numbers, and social media such as LinkedIn after their informed consent through written message requests, which clearly stated the purpose of the study. Convenience sampling is a non-probabilistic technique often used in quantitative studies, where subjects are selected based on accessibility to the researcher (Suen et al., 2014). According to Bryman (2016), convenience sampling can be used when well-defined populations and sampling frames are not available. While it can be applied in both qualitative and quantitative research, it lacks generalizability to the target population (Obilor & Isaac, 2023).

3.1. Measurement

The measurement scales used in this study were adapted from previous research. The employee performance scale, comprising 12 items, was derived from Kamal et al. (2016), who reported a Cronbach's alpha of 0.911, indicating an acceptable level of reliability. In addition, the shorter version of the technostress scale was taken from Saleem et al. (2021), which was first created by Tarafdar et al. (2007), and it showed a Cronbach's alpha of 0.946 in their study. The WFH scale, consisting of seven items, was adapted from Ishak et al. (2022) and demonstrated a Cronbach's alpha of 0.694. Finally, the digital literacy scale was sourced from the study by Mohd Abas et al. (2019), which reported a Cronbach's alpha of 0.985. The response format employed in this study was a Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree).

However, the current study verified the reliability and validity of the scale using Cronbach's alpha, AVE, composite reliability, and HTMT

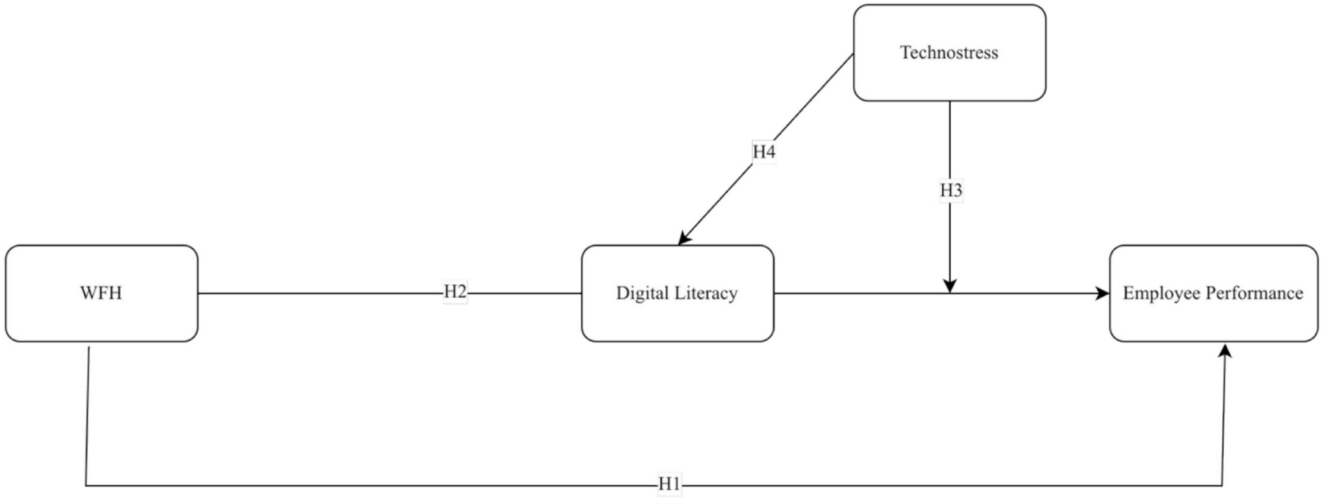


Fig. 1. Framework of the research.

ratios; all these values met the set threshold, and the results are presented in Tables 3 and 4.

3.2. Non-response biased test

Non-response bias can occur when actual survey respondents differ from sampled respondents. There are multiple methods to test for non-response bias; however, this study adopted the comparison of early to late respondents, which is widely accepted by previous studies (Wagner & Kemmerling, 2010). This study used Levene's homogeneity of variance test with SPSS software. In respect of the explanation of the findings, if results found significant that non-response bias exists, on the other side, results revealed the not significant findings among groups, which warrants that there is no non-response bias in this study. In simpler terms, the two groups (early and late responses) did not show significant differences. This study, as shown in Table 1, shows the insignificant findings of all four constructs, meaning that all variables are not significantly different. This study concludes that there is no significant difference between the respondents who answered and those who did not, indicating that further data collection from the target population is not necessary.

3.3. Common method bias (CMB)

Common method bias, also known as common method variance (CMV), can occur in survey-based research when data are collected from a single source at a single point in time (Podsakoff et al., 2003). The issue of CMB can inflate or deflate the proposed relationships among variables. Therefore, Herman's single-factor test was used to evaluate CMB (Podsakoff et al., 2012). This study employed SPSS to run Herman's single-factor test using factor analysis for one factor. Therefore, according to Table 2, the total variance of 23.231% is below the 50% threshold, indicating that CMB is not a dominant issue. Furthermore, the present study tested the full collinearity (VIF) of the proposed variables to provide a rigorous assessment, as recommended by Kock (2015). The results in Table 3 show that all VIF values are recorded under the

Table 1
Non-response biased test results.

Independent samples test											
		Levene's test for equality of variances		*t*-Test for equality of means							
		F	Sig.	t	df	Significance		Mean difference	Std. error difference	95% confidence interval of the difference	
						One-sided p	Two-sided p			Lower	Upper
Technostress	Equal variances assumed	0.839	0.361	−0.933	317	0.176	0.351	−0.32916	0.35269	−1.02307	0.36474
	Equal variances not assumed			−0.701	4.071	0.261	0.521	−0.32916	0.46936	−1.62339	0.96507
Digital literacy	Equal variances assumed	0.009	0.926	0.048	317	0.481	0.961	0.01302	0.26864	−0.51553	0.54156
	Equal variances not assumed			0.046	4.117	0.483	0.965	0.01302	0.28127	−0.75926	0.78530
WFH	Equal variances assumed	0.321	0.571	0.385	317	0.350	0.701	0.12575	0.32676	−0.51714	0.76865
	Equal variances not assumed			0.481	4.205	0.327	0.654	0.12575	0.26117	−0.58567	0.83717
Employee performance	Equal variances assumed	0.008	0.928	−0.139	317	0.445	0.889	−0.05016	0.36008	−0.75861	0.65829
	Equal variances not assumed			−0.132	4.115	0.450	0.901	−0.05016	0.37907	−1.09109	0.99078

Table 2
Common method bias results.

Total variance explained						
Component	Initial Eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	13.474	23.231	23.231	13.474	23.231	23.231
2	6.864	11.835	35.066			
3	5.533	9.540	44.606			
4	3.115	5.371	49.977			
5	2.055	3.544	53.520			
6	1.829	3.153	56.673			
7	1.528	2.635	59.308			
8	1.411	2.432	61.741			
9	1.316	2.269	64.010			
10	1.205	2.077	66.087			
11	0.944	1.627	67.714			
12	0.924	1.593	69.307			
13	0.824	1.421	70.728			
14	0.778	1.342	72.070			
15	0.753	1.298	73.368			
16	0.732	1.261	74.629			
17	0.716	1.234	75.863			
18	0.658	1.134	76.998			
19	0.644	1.110	78.108			
20	0.635	1.096	79.203			
21	0.592	1.021	80.225			
22	0.570	0.982	81.207			
23	0.542	0.934	82.141			
24	0.530	0.914	83.055			
25	0.493	0.850	83.904			
26	0.487	0.840	84.744			
27	0.468	0.807	85.551			
28	0.456	0.786	86.337			
29	0.435	0.750	87.087			
30	0.409	0.706	87.792			
31	0.407	0.702	88.495			
32	0.385	0.663	89.158			
33	0.370	0.637	89.796			
34	0.359	0.619	90.415			
35	0.355	0.611	91.026			
36	0.337	0.582	91.608			
37	0.320	0.552	92.160			
38	0.318	0.548	92.708			
39	0.299	0.515	93.223			
40	0.290	0.500	93.723			
41	0.283	0.488	94.211			
42	0.277	0.478	94.689			
43	0.255	0.440	95.130			
44	0.247	0.426	95.555			
45	0.241	0.415	95.971			
46	0.228	0.393	96.364			
47	0.224	0.385	96.749			
48	0.220	0.379	97.128			
49	0.200	0.345	97.473			
50	0.191	0.329	97.802			
51	0.189	0.326	98.127			
52	0.184	0.318	98.445			
53	0.176	0.304	98.749			
54	0.166	0.287	99.036			
55	0.162	0.280	99.316			
56	0.152	0.262	99.578			
57	0.130	0.224	99.802			
58	0.115	0.198	100.000			

Extraction Method: Principal Component Analysis.

Table 3
VIF.

Model[a]		Collinearity statistics	
		Tolerance	VIF
1	Technostress	0.968	1.033
	Digital Literacy	0.464	2.157
	WFH	0.454	2.204

[a] Dependent Variable: Employee performance.

threshold, which indicates that CMB is unlikely to be significantly biased in this study.

In simple words, although CMB cannot be entirely ruled out from the study, the above statistical assessment suggests that it is unlikely to pose a serious threat to the validity of the study. Therefore, the data were considered appropriate for subsequent analyses.

4. Data analysis and findings

SPSS and SmartPLS-4 (PLS-SEM) were used to evaluate the collected data and draw conclusions. The PLS-SEM was run in two stages: measurement and structural model. The measurement model evaluated the construct's reliability and validity, and the structural model evaluated the relationship between constructs and accepted or rejected the proposed hypothesis.

4.1. Measurement model

Cronbach's alpha coefficients were calculated to assess the internal consistency of the constructs, with a threshold of 0.70 indicating acceptable reliability. The results of this study meet this criterion. Composite reliability was assessed using two alternative measures: rho_A and rho_C. The composite reliability values (rho_A and rho_C) exceeded the threshold of 0.70 for all constructs, indicating that their reliability was achieved. Additionally, the average variance extracted (AVE) was calculated to assess the constructs' convergent validity. Specifically, the AVE values should exceed 0.50, and the recorded AVE values for each construct were above this threshold. Thus, the convergent validity of the constructs was confirmed. Table 4 presents the reliability and validity measures of the constructs.

Furthermore, the Heterotrait-Monotrait Ratio (HTMT) criterion was used to assess the discriminant validity of the constructs, as recommended by Henseler et al. (2015). The HTMT is a robust criterion that evaluates discriminant validity by comparing the correlations between indicators across different constructs with the correlations within the same construct. According to Henseler et al. (2015), discriminant validity is established when the HTMT values are recorded below 0.85 as per the strict criterion. As per Table 5, all HTMT values lie between 0.104 and 0.674, which are below the threshold of 0.85. Thus, the HTMT finding confirms that all constructs are empirically different from each other, thereby establishing adequate discriminant validity for the measurement model.

4.2. Structural model

The structural model aims to explain the causal or predictive relationships among the study variables. To assess the path coefficient, SmartPLS provides t-statistics; the value of t-statistics should be greater than 1.96 at a significance level of 0.05 (Hair et al., 2019). Hence, as per Table 6 results, hypotheses H2 and H3, except H1, recorded greater t-statistics, 2.708 and 2.379, respectively, at the 0.007 and 0.018 significance levels. Thus, H2 and H3 are supported, and H1 is rejected.

Furthermore, the PROCESS macro was run on smartPLS-4 to test the 2nd-stage moderated mediation to verify Hypothesis 4 (Hair, 2014; Hayes, 2017). Table 7 and the moderated-mediation graph in Fig. 2 show the conditional indirect effect and its level of confidence (CI) within the limit, meaning no zero is involved, which accepts hypothesis 4.

5. Discussion of the findings

The direct relationship between WFH and employee performance was also examined. As per Table 3, the value of $t = -0.812$, $p = 0.417$ shows the insignificant relationship between WFH and employee performance at public universities in Malaysia. Moreover, the mediating role of digital literacy between WFH and employee performance was examined. The findings in Table 3 show the positive and significant mediating role of digital literacy between WFH and employee performance. The type of mediation is full mediation; in other words, when there is full mediation, the influence of WFH on employee performance disappears entirely once the digital literacy mediating variable is considered. Employees become more adept at utilizing digital tools and platforms and are better positioned to fulfill their job responsibilities effectively. Through the intervention of digital skills, remote work provides work engagement flexibility, which leads to higher employee performance.

The moderating role of technology stress in the relationship between digital literacy and employee performance was also examined. The results show the negative and significant ($t = -2.379$, $p = 0.018$) moderating role of technostress in the relationship between digital literacy and employee performance. In other words, the negative moderating role of technostress weakens the relationship between digital literacy and employee performance in Malaysian public universities. Although digital literacy is positively associated with employee performance, the presence of technostress dampens this relationship. Technostress, which arises from the continuous use of technology, can hinder the positive effects of digital literacy on performance by causing anxiety, burnout, or resistance to technology adoption among employees.

Moreover, the moderating role of technostress on the indirect effect of WFH on employee performance through digital literacy was tested. Table 6 shows the significant negative moderated mediation interaction. Regarding the level of moderated mediation, the indirect relationship was strong when technostress was low (effect = 0.193, CI [0.127, 0.259]) and moderate at the mean level (effect = 0.127, CI [0.077, 0.177]). Finally, when technostress was high, the indirect relationship was the weakest (effect 0.061, CI [0.002, 0.120]). In simple words, the higher the level of technostress, the more diminished the effect of digital literacy on the benefits of WFH to improve the lecturer's performance.

5.1. Implications of the study

The implications related to WFH, digital literacy, and technostress are important not only for universities but also for individuals. Theoretically, this is the first study to investigate the direct relationship between WFH and employee performance in Malaysian public universities, with the mediating role of digital literacy and the moderating role of technostress. The indirect effect of WFH on employee performance through digital literacy intervention advises organizations to implement flexible working hours to achieve the desired WFH outcomes, such as employee satisfaction and cost savings. It is also suggested that organizations, particularly universities, should consider the effective and efficient implementation of WFH policies for longer gain.

Additionally, digital literacy is very important for higher employee performance; hence, organizations such as universities should prioritize providing training and resources to enhance employees' digital skills, such as workshops and online courses, and empower employees to use digital tools, communicate online, and manage their tasks remotely. Moreover, universities should put their efforts into develop a helpful technological environment that reduces the stress caused by continual connectivity and information overload. Promoting responsible technology use and providing tools for controlling technostress can enhance

Table 4
Reliability of constructs.

Constructs	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
Digital Literacy	0.904	0.908	0.919	0.510
Employee performance	0.929	0.939	0.939	0.566
Technostress	0.896	0.923	0.909	0.559
WFH	0.763	0.769	0.850	0.587

Table 5
Validity of constructs.

Constructs	Digital Literacy	Employee performance	Technostress	WFH	Technostress × Digital Literacy
Digital Literacy					
Employee performance	0.284				
Technostress	0.214	0.249			
WFH	0.764	0.279	0.255		
Technostress × Digital Literacy	0.104	0.11	0.255	0.136	

Table 6
Hypothesis findings.

Hypothesis	Beta	Standard deviation	t-Statistics	*p*-Values
H1: WFH - > Employee performance	0.054	0.066	0.812	0.417
H2: WFH - > Digital Literacy - > Employee performance	0.109	0.040	2.708	0.007
H3: Technostress x Digital Literacy - > Employee performance	−0.100	0.042	2.379	0.018

Table 7
Results of conditional indirect effect.

Moderated indirect relationship	Effect	Standard error	Lower limit CI	Upper limit CI
ST X WFH-DL-EP	−0.074	0.043	−0.143	−0.004

Level of moderated indirect relationship	Effect	Standard error	Lower limit CI	Upper limit CI
WFH - > DL - > EP conditional on EP at +1 SD (+1.365)	0.061	0.059	0.002	0.120
WFH - > DL - > EP conditional on EP at −1 SD (−1.365)	0.193	0.066	0.127	0.259
WFH - > DL - > EP conditional on EP at Mean (0)	0.127	0.050	0.077	0.177

Bootstrap Process sample size: 5000, CI: confidence interval.
Note: Work-from-home (WFH), Digital literacy (DL), Technostress (ST), Employee performance (EP).

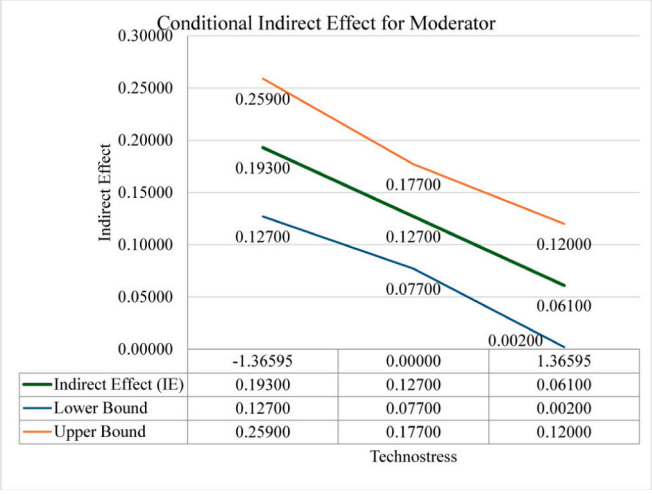


Fig. 2. Conditional indirect effect for moderation.

employee performance and well-being. Thus, considering the above implications, universities can improve their employees' performance, satisfaction, and overall success in a digital-centric work landscape.

Finally, this study contributes to the Transaction Model of Stress and Coping Theory (TMSC), which examines WFH, employee performance, digital literacy, and technostress. This finding potentially contributes to the existing knowledge of TMSC, which will be useful for academicians and policymakers.

5.2. Limitations and future recommendations

The current study is limited to public universities; future studies can consider private universities for better generalization of the results. This raises concerns about the population's representation, as employees from different institutions may have divergent perceptions and consequently generate different results. Moreover, the results obtained may not be generalized to employees in different fields at public universities, as the research has focused primarily on the education sector. For instance, lecturers who adapted to situations had more exposure to the use of advanced technology than the staff in private. This study also reveals a gap in technological skills among staff members at public and private universities. Future research can be more interesting if it expands the samples used, not just public university employees. Future studies may also broaden the scope of the study beyond universities in Malaysia to increase the accuracy of the results, where an overall understanding of the populations can be achieved. The current study adopted a non-probability, convenience sampling strategy due to the unavailability of a defined sampling frame, reliable population parameters, and time constraints; hence, future studies can consider a probability-based sampling strategy, such as random or stratified sampling, to enhance representativeness and generalizability.

The lack of relative factors has led to limitations in this study. To improve outcomes, more relative factors influencing employee performance should be included in future research, such as e-training and adaptive expertise. The cross-sectional approach has been used for the current study; future studies can collect data using a longitudinal approach to generalize the findings.

5.3. Conclusion of the study

This study's findings revealed the perfect mediating role of digital literacy between WFH and employee performance. In terms of moderation, this study's findings showed the negative moderating role of technostress on the relationship between digital literacy and employee performance in public universities in Malaysia. This study also found that technostress moderated the indirect effect between WFH and employee performance through digital literacy.

The WFH direct relationship was found to be insignificant. Anyway, it has an influential effect on employee performance through conditioning to digital literacy. The mediating role of digital literacy enhances employees' abilities to use digital tools, so these abilities give leverage to employees to take advantage of WFH to improve their performance. Additionally, the moderating role of technostress is playing a negative role in weakening the relationship between digital literacy and employee performance. In simple words, high levels of technostress seem to struggle to effectively translate their digital literacy into improved performance due to the stress associated with technology. Thus, in the case of technostress, universities should provide training to neutralize the employees' stress regarding technology use and allow

them to fully utilize their digital skills.

CRediT authorship contribution statement

Dong Pengfei: Writing – original draft, Funding acquisition, Conceptualization. **Muhammad Imran:** Writing – review & editing, Supervision, Software, Methodology, Formal analysis. **A.K. Mahbubul Hye:** Writing – review & editing, Validation, Resources, Methodology.

Ethical statement

This study was reviewed and approved by the Research Ethics Committee of the School of International Education, Zhengzhou University of Industrial Technology (Approval No.: ZZUIT-SIE/ETH/2025/1002). All participants were informed about the objectives and procedures of the research before participation, and their informed consent was obtained prior to data collection. Participation was voluntary, and respondents were assured of anonymity and confidentiality of their responses. The study was conducted in accordance with the ethical standards of the Declaration of Helsinki and the institutional guidelines of Zhengzhou University of Industrial Technology, China.

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Declaration of competing interest

The authors declare no conflict of interest related to this study. This research was conducted independently, and no financial or non-financial relationships influenced the study design, data collection, or analysis.

Data availability

Data will be made available on request.

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